PAPER_884: Net Energy E^+/E^- Evolution

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** NetEnergyEplusEminusEvolutionCalc (CP4 #468) **CVW:** v2.0.0 compliant

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Abstract

Net energy evolution $E\_net(t) = E^+(t) + E^-(t) = E_0 \cdot \exp(\kappa t + [SSq] \cdot t/26) \cdot S_{26} \cdot (2R-1)$. The sign is determined by the buoyancy ratio R: R>0.5 yields expanding universe, R<0.5 yields eroding/contracting regime, R=0.5 is perfectly balanced. Verifies the identity $E^+ + E^- \equiv E\_net$.

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1. Core Equations

$$E\_net = E^+ + E^- = E_0 \cdot \exp(...) \cdot S_{26} \cdot (2R-1)$$

verified identity

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2. Parameters

Parameter	Default	Description
E_0	1.0 J	Initial energy scale
t	0.0 s	Time parameter
F_UBi_over_FU	0.8	Buoyancy-to-field ratio

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3. UQFF Integration

This calculator integrates negative_et_erosion.py from Session 205 into the CP4 pipeline as class #468.

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